

## Functions

Simple mean function

```
lazy.mean <- function(x) {  
  sum(x) / length(x)  
}
```

```
x <- 1:30
```

## optim

x is the data set

## Define log likelihood function

```
ll.exp <- function(lambda, x) {  
  n <- length(x)  
  n * log(lambda) - lambda * sum(x)  
}
```

## Define dataset

```
set.seed(400)  
data <- rexp(n = 10000, rate = 3)
```

## Find MLE

par = starting vlaue(s) for parameter(s)

```
optim(  
  par = 5, # starting parameter  
  fn = ll.exp,  
  method = "L-BFGS-B", # optimization method  
  lower = 1e-6, # lower bound  
  control = list(fnscale = -1), # maximize instead of minimize  
  x = data  
)
```

```
## $par  
## [1] 3.008847  
##  
## $value  
## [1] 1015.568  
##  
## $counts  
## function gradient  
##      10      10  
##  
## $convergence  
## [1] 0  
##  
## $message  
## [1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```